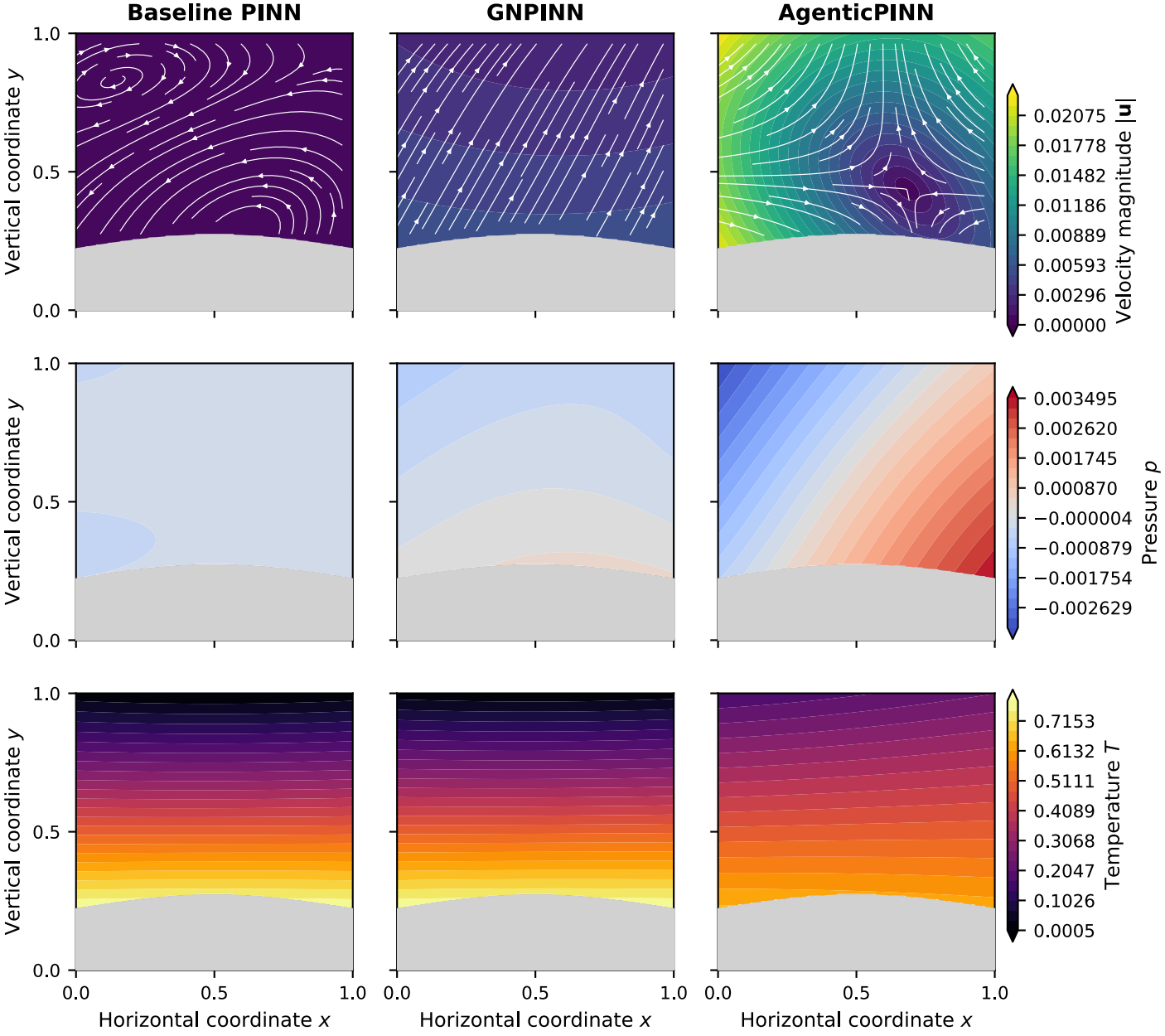

Simplified crystal-growth fields at $t = 0.20$
Baseline PINN
GNPINN
AgenticPINN
Vertical coordinate y
Horizontal coordinate x
Velocity magnitude $|\mathbf{u}|$
0.02075
0.01778
0.01482
0.01186
0.00889
0.00593
0.00296
0.00000
Pressure p
0.003495
0.002620
0.001745
0.000870
−0.000004
−0.000879
−0.001754
−0.002629
Temperature T
0.7153
0.6132
0.5111
0.4089
0.3068
0.2047
0.1026
0.0005
0.0
0.5
1.0